

AfyaPlus Generative AI Observability - Executive Summary

To: CTO and Medical Director | From: Applied AI Engineering Team

Executive Summary

AfyaPlus clinical AI ran six weeks without automated evaluation. This week we deployed evaluation, drift monitoring, and cost tracking. gpt-4o-mini averaged ROUGE-L 0.142 vs gpt-4o 0.179. Monthly spend reached \$710 (47% of cap).

Quality Performance Breakdown

Across 15 clinical questions, automated gate pass rate is 0% (mini) and 0% (gpt-4o) due to BLEU/ROUGE formatting gaps. LLM judge safety scores are stronger: mini 3.953 (93% above 3.0 threshold), gpt-4o 3.98 (93% above 3.0). Triage judge groundedness averaged 3.467 (mini).

Cost and Efficiency Analysis

30-day projected spend: \$710 at 75/25 mini/gpt-4o split. Routing low-acuity traffic to mini where judge scores clear the 3.0 safety bar saves \$542/month. Human-in-the-loop review costs ~\$12 per case vs ~\$0.002 per automated inference.

Systemic Operational Risks

ROUGE-L drifted from 0.425 (Month 1) to 0.3013 (Month 3). First alert: Month 1 column rouge_l. Latency rose to 1096ms avg - hallucination risk increases when quality scores drop below gate thresholds.

Actionable Engineering Roadmap

1) Route USSD and Mobile App triage to gpt-4o-mini (saves ~\$542/mo; judge scores avg 4.0). 2) Deploy monthly Evidently drift alerts to Slack when ROUGE-L drops >10%. 3) Reserve gpt-4o for Web Portal high-acuity cases only.